

Fundamental Limits of Perfect Concept Erasure

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Abstract

Concept erasure is the task of erasing information about a concept (e.g., gender or race) from a representation set while retaining the maximum possible utility — information from original representations. Concept erasure is useful in several applications, such as removing sensitive concepts to achieve fairness and interpreting the impact of specific concepts on a model’s performance. Previous concept erasure techniques have prioritized robustly erasing concepts over retaining the utility of the resultant representations. However, there seems to be an inherent tradeoff between erasure and retaining utility, making it unclear how to achieve perfect concept erasure while maintaining high utility. In this paper, we offer a fresh perspective toward solving this problem by quantifying the fundamental limits of concept erasure through an information-theoretic lens. Using these results, we investigate constraints on the data distribution and the erasure functions required to achieve the limits of perfect concept erasure. Empirically, we show that the derived erasure functions achieve the optimal theoretical bounds. Additionally, we show that our approach outperforms existing methods on a range of synthetic and real-world datasets using GPT-4 representations.

supervised manner have showcased an impressive array of capabilities across a range of applications (Ahn et al., 2022; Sun, 2023; Singhal et al., 2023). Owing to their success, in many applications practitioners rely on data representations from large models (Kenton and Toutanova, 2019; Dosovitskiy et al., 2021; Radford et al., 2021; Touvron et al., 2023) and fine-tune smaller networks on top of these representations. Such representations often encode a diverse set of *concepts* (Hinton et al., 1986) about the underlying data. For example, concepts associated with a video representation can be the geographic location or pose of a person. Concepts can also be more abstract, e.g., text representations may encode writing style (Patel et al., 2023), topics (Sarkar et al., 2023), or even sensitive information like gender or race of the author (Bolukbasi et al., 2016). Since practitioners cannot control the concepts encoded in representations, machine learning models may learn to rely on such sensitive concepts, thereby inadvertently impacting downstream applications. For example, while developing a resume screening system, an organization may want to ensure that the candidate’s gender does not affect the hiring decisions. This calls for techniques that enable practitioners to erase certain concepts encoded in data representations.

To address these challenges, we focus on *concept erasure* (Ravfogel et al., 2020, 2022a; Chowdhury et al., 2024), i.e., removing information about a concept (e.g., gender) from a representation set. This is performed by transforming the original representations using an erasure function. The transformed representations should encode minimal information about the erased concept (*privacy*) while retaining the maximal information about the original representations (*utility*). This differs from conventional invariant representation learning (Zhao et al., 2019; Nguyen et al., 2021; Zhao et al., 2022; Sadeghi et al., 2022), which only retains information about a specific task (e.g., hiring). Since concept erasure is task-agnostic, its output representations can be used in a wide range of applications. This is useful when a data distributor wants to share data represen-

1 INTRODUCTION

Modern large-scale machine learning models (Achiam et al., 2023; Team et al., 2023) trained in a self-

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tations with its customers while safeguarding sensitive concepts, such as the gender or race of the data producer (Zemel et al., 2013). Additionally, prior work has shown that concept erasure is useful in improving fairness (Ravfogel et al., 2020; Chowdhury and Chaturvedi, 2022; Belrose et al., 2024) and interpretability (Gonen et al., 2020; Elazar et al., 2021).

A long line of work (Bolukbasi et al., 2016; Ravfogel et al., 2020, 2022a,b; Chowdhury and Chaturvedi, 2022; Chowdhury et al., 2024) has focused on developing effective concept erasure techniques. Most of these approaches prioritize only erasing concept information. However, in concept erasure, we want to achieve perfect erasure while retaining maximum utility. Motivated by this challenge, we aim to answer the research question: *What is the maximum utility that can be retained while perfectly erasing a concept, and what functions achieve perfect erasure?*

In this work, we take a step towards answering the above question by deriving the functions that achieve perfect concept erasure. We formalize the concept erasure task using information-theoretic tools, introducing the definitions of perfect and relaxed concept erasure. We borrow tools from privacy and make a conceptual connection with trade-offs between utility and privacy (Sreekumar and Gündüz, 2019; Bertran et al., 2019; Razeghi et al., 2020; Atashin et al., 2021) and build on (Calmon et al., 2015, 2017) that characterize the fundamental limits of perfect privacy. In these data settings, we derive the erasure functions that achieve perfect concept erasure, which we call *perfect erasure functions* (PEFs). In controlled settings, we show that existing approaches either reveal a significant amount of concept information or entirely lose all utility information from the original representations.

To summarize, our primary contributions are:

- We formalize the task of concept erasure, introducing the definitions of perfect and relaxed concept erasure using information-theoretic tools.
- We hypothesize constraints on the underlying data required to achieve the information-theoretic outer bounds for perfect concept erasure.
- We derive the perfect erasure functions under different data conditions.
- We extensively evaluate PEF on synthetic and real-world datasets, including GPT-4 representations.

2 BACKGROUND

Concept Erasure. The idea of removing information from data representations was motivated by the problem of removing gender information from GloVe embeddings (Bolukbasi et al., 2016). This task was

later termed as *concept erasure* (Ravfogel et al., 2020, 2022a), which involved removing information about a *concept* (a random variable) from a representation set. Concept erasure differs from conventional approaches to removing information from representations, such as adversarial learning (Zemel et al., 2013; Madras et al., 2018; Elazar and Goldberg, 2018; Sadeghi et al., 2022; Iwasawa et al., 2018; Chowdhury et al., 2021; Xie et al., 2017), which aim to learn invariant representations for a specific task. For concept erasure, the user does not have access to a specific task and should retain the maximum utility from original representations.

A simplified form of this task is *linear concept erasure* (Ravfogel et al., 2020; Dev et al., 2021; Ravfogel et al., 2022a), which removes concept information in a manner that prevents a linear adversary (e.g., a linear classifier) from extracting it. Initial approaches involve projecting representations onto the nullspace of linear concept classifiers, either once (Bolukbasi et al., 2016; Haghhighatkhah et al., 2022) or in an iterative manner (Ravfogel et al., 2020). Subsequent work (Ravfogel et al., 2022a) generalized the iterative nullspace projection objective as a minimax game. Recently, Belrose et al. (2024) show that a concept is perfectly linearly erased only when different concept groups share the same centroid representation and provide an optimal algorithm to achieve this condition.

Though theoretically sound and effective in practice, linear concept erasure has limitations, as the erased concept can still be retrieved using a non-linear adversary. Recent works have focused on performing *general concept erasure*, which tries to protect the concept against an arbitrarily strong adversary. One approach, utilized by (Ravfogel et al., 2022b; Shao et al., 2023), involves projecting inputs onto a non-linear space and then applying linear concept erasure techniques. On the other hand, Chowdhury and Chaturvedi (2022); Chowdhury et al. (2024) learn a parameterized erasure function using rate-distortion based objectives. While empirically promising, these approaches cannot theoretically guarantee perfect erasure nor the retention of maximum possible utility.

General concept erasure techniques often showcase a significant information/utility loss (Chowdhury et al., 2024), which is expected as perfectly erasing a concept may remove other information. An extreme case occurs when a function outputs random representations, which perfectly erases a concept but results in complete information loss (Lowy et al., 2022). Therefore, we quantify the maximum possible information that can be retained while perfectly erasing a concept. In this work, we use mutual information to quantify the utility and privacy during concept erasure (Sankar et al., 2013). Mutual information minimization has been used to improve

privacy (Duchi et al., 2014; Bertran et al., 2019) and fairness (Mary et al., 2019; Lowy et al., 2022) in many applications. However, we do not optimize the mutual information terms directly; instead, we utilize their outer bounds. This problem can be studied using other divergence measures as well. We use mutual information as its outer bounds enable us to derive the analytic form of the erasure functions.

A long line of work in information theory (Calmon et al., 2017; Sreekumar and Gündüz, 2019; Bertran et al., 2019; Atashin et al., 2021) has focused on achieving an optimal tradeoff between privacy, utility, and function complexity by optimizing the associated mutual information terms. Bottleneck CLUB (Razeghi et al., 2023) provides a general framework to unify these approaches. Concept erasure is a special case of this framework, where we focus only on utility and privacy (also called the privacy funnel (Makhdoumi et al., 2014)). Calmon et al. (2015) was the first to study perfect privacy (or perfect erasure), deriving the PIC conditions necessary to achieve it within the privacy funnel framework. Calmon et al. (2017) derived the mutual information bounds for the privacy funnel setting, while Hsu et al. (2018) generalized the privacy funnel problem to a broader class of f -divergences. Building on this, Wang and Calmon (2017); Wang et al. (2019) focused on erasing functions of the data such that those cannot be reconstructed (using MMSE) and derived sharp privacy-utility tradeoff bounds using χ^2 -divergence. More recent works (Razeghi et al., 2024; Huang and El Gamal, 2024), leveraged variational approximation using deep networks to optimize the privacy funnel. In contrast to these approaches, we do not optimize the mutual information terms. Instead, we build on the theoretical framework presented by Calmon et al. (2017). Using the mutual information outer bounds from (Calmon et al., 2017), we derive the data constraints and erasure functions necessary for achieving perfect concept erasure. Next, we discuss the minimum entropy coupling, which we incorporate into the erasure functions.

Minimum Entropy Coupling (MEC). Minimum entropy coupling Γ is a joint distribution over m input probability distributions $(p_1, \dots, p_m)$, with the minimum entropy. In our work, we are only interested in finding a coupling between two distributions, $\Gamma(p, q)$.

$$\Gamma_{\min}(p, q) = \arg \min_{\Gamma} H_{\Gamma}(p, q) = -\Gamma_{ij} \log \Gamma_{ij},$$

where $\sum_j \Gamma_{ij} = p_i, \sum_i \Gamma_{ij} = q_j$. It is easy to see that the minimum entropy, $H_{\min}(p, q)$, achieved by MEC, $\Gamma_{\min}(p, q)$, also maximizes the mutual information: $I(p; q) = H(p) + H(q) - H_{\min}(p, q)$, between two distributions p and q . However, computing the minimum entropy coupling is NP-Hard (Kovačević et al., 2015).

Later, Kocaoglu et al. (2017a) introduced a greedy algorithm to efficiently approximate the MEC. Subsequent works (Kocaoglu et al., 2017b; Rossi, 2019; Compton et al., 2022) showed that the approximation produced by the greedy algorithm is tight. Specifically, while computing the MEC between two distributions, the greedy approach produces approximations within 0.53 bits (Compton et al., 2023). We use the greedy algorithm for efficiently computing the MEC to derive the perfect erasure functions. In the following sections, we will formalize concept erasure, quantify the outer bounds, and derive perfect erasure functions.

3 CONCEPT ERASURE

3.1 Problem Statement

Concept erasure tries to erase the effect of a concept, A (a random variable with support $\mathcal{A}$), from a representation set, X (random variable with support $\mathcal{X}$). This is usually done by learning a transformation of the original representations, $Z = f(X)$ (random variable with support $\mathcal{Z}$). We will refer to the transformation map, f , as the *erasure function*. In our setup, the erasure function f only has access to X and not A . This allows f to generalize to examples where the concept A is not available. Note that although prior works have utilized various erasure techniques, they can all be generalized using an erasure function, f . For example, linear concept erasure approaches (Bolukbasi et al., 2016; Ravfogel et al., 2020) use a product of projection matrices as $f(x) = \prod_j \mathbf{P}_j x$, kernelized linear erasure (Ravfogel et al., 2022b) use $f(x) = \prod_j \mathbf{P}_j \Phi(x)$ (where $\Phi(\cdot)$ is non-linear), Chowdhury and Chaturvedi (2022); Chowdhury et al. (2024); Huang and El Gamal (2024) explicitly parameterize f using neural networks.

Assumptions. We make the following assumptions in our concept erasure setup:

- (A1) The Markov property: $A \longrightarrow X \xrightarrow{f} Z$, which implies $I(Z; A|X) = 0$.
- (A2) The support of the input and output representation sets, $\mathcal{X}$ and $\mathcal{Z}$, are finite.
- (A3) The support of the concept set is finite, $\mathcal{A} = \{a_1, a_2, \dots, a_{|\mathcal{A}|}\}$.
- (A4) The representations for each concept group are sampled from the distribution, $\forall i \in \{1, \dots, |\mathcal{A}|\}$, $P_i = P(X|A = a_i)$. For all pairs of distributions (P_i, P_j) with $i \neq j$, their supports (defined as $\mathcal{X}_i = \text{supp}(P_i)$) are disjoint, i.e., $\mathcal{X}_i \cap \mathcal{X}_j = \emptyset$. Similarly, the output representations for each concept group is $P(Z|A = a_i)$.
- (A5) The size of the input representation support is larger than the size of concept support, $|\mathcal{X}| > |\mathcal{A}|$.

An example of such a setup is erasing gender information from English word embeddings. The representation set is finite since the number of English words is finite. Therefore, even though data representations are typically continuous, assuming a categorical distribution with a large support set is reasonable.

Objective. The primary objective of concept erasure is to construct an erasure function, f , such that these two properties: (a) effectively erase a concept, A , achieving $I(Z; A) \approx 0$ and (b) retaining information about the original representations, X , achieving high $I(Z; X)$. The resultant objective is (for a small constant $\epsilon > 0$):

$$\max_f I(Z; X) \text{ subject to } I(Z; A) \leq \epsilon, \quad (1)$$

where $Z = f(X)$. The concept erasure objective shown above is reminiscent of the information bottleneck objective (Tishby et al., 2000), which focuses on achieving a good tradeoff between $I(Z; A)$ and $I(Z; X)$. In concept erasure, the objective is to robustly erase concept A , resulting in near-zero $I(Z; A) \approx 0$, even if it is at the expense of a reduced utility, $I(Z; X)$.

3.2 Types of Concept Erasure

We note that optimizing the objective in Eq. 1 is difficult, as estimating mutual information using a finite number of samples is challenging (Song and Ermon, 2019; McAllester and Stratos, 2020). Rather than optimizing the mutual information terms, we use their outer bounds to derive the erasure functions. Next, we introduce several definitions involving concept erasure.

Definition 3.1 (Perfect Concept Erasure). *An erasure function, f , achieves perfect concept erasure if $I(Z; A) = 0$, where $Z = f(X)$.*

Perfect concept erasure ensures that no information about the concept is present in the resultant representations, $I(Z; A) = 0$. However, achieving perfect erasure may not always be feasible. Our next result provides the conditions under which perfect concept erasure is feasible. For deriving the conditions of perfect erasure, we rely on the notion of principal inertia components (PICs) (Calmon et al., 2017). Intuitively, PICs capture the correlation between two random variables X and A , and can be viewed as a decomposition of the joint distribution, $p_{A,X}$ (see details in Appendix A.1). Note that all of the results in this paper are derived under Assumptions (A1)-(A5) unless otherwise stated.

Lemma 3.2 (Perfect Concept Erasure Achievability). *Under the assumptions (A1)-(A4), for a joint distribution $p_{A,X}$ with finite support defined over $A \times \mathcal{X}$, perfect concept erasure, $I(Z; A) = 0$, is achievable if either (a) smallest principal inertia component, $\lambda_d(A, X) = 0$ or (b) $|\mathcal{X}| > |\mathcal{A}|$.*

This lemma states that perfect erasure is feasible when either the smallest PIC is zero, $\lambda_d(A, X) = 0$, or $|\mathcal{X}| > |\mathcal{A}|$ (proof in Appendix A.1). Throughout the paper, we assume (A5): $|\mathcal{X}| > |\mathcal{A}|$, which means perfect erasure is always feasible irrespective of $\lambda_d(A, X)$. This is a reasonable assumption in practice as the size of the representation space is typically much larger than the number of concepts.

Note that Definition 3.1 for perfect concept erasure only requires the concept to be erased from learned representations, $I(Z; A) = 0$. It does not impose any constraints on the utility of the learned representations, $I(Z; X)$. Since concept erasure also involves retaining maximum utility, it is natural to question the maximum achievable utility $I(Z; X)$ during perfect erasure.

Lemma 3.3 (Perfect Erasure Bound). *Under the assumptions (A1)-(A5), if an erasure function $f: \mathcal{X} \rightarrow \mathcal{Z}$ achieves perfect concept erasure $I(Z; A) = 0$, then the utility $I(Z; X)$ is bounded as:*

$$I(Z; X) = I(Z; X|A) \leq H(X|A),$$

where the equality is satisfied when $H(X|Z, A) = 0$.

The above result presents the outer bound for the utility, $I(Z; X)$, during perfect concept erasure. The complete proof is in Appendix A.2. Next, we focus on a slightly different setting of concept erasure where the privacy leakage about the concept can be non-zero, $I(Z; A) = \epsilon$ ($\epsilon > 0$). Specifically, we are interested in understanding the best achievable privacy (minimum $I(Z; A)$) for any utility u , such that $I(Z; X) \geq u$. To address this question, we present the notion of an *erasure funnel*¹ introduced in information theory literature.

Definition 3.4 (Erasure Funnel (Calmon et al., 2017)). *For $0 \leq u \leq H(X)$, representations $Z = f(X)$, and a fixed joint distribution over $A \times \mathcal{X}$, we define the erasure funnel function as:*

$$\epsilon(u) = \inf_f \{I(Z; A) | I(Z; X) \geq u, A \rightarrow X \xrightarrow{f} Z\}. \quad (2)$$

We use the erasure funnel to define the setting where concept leakage is non-zero, $\epsilon(u) > 0$. We term this setting *relaxed concept erasure*.

Definition 3.5 (Relaxed Concept Erasure). *An erasure function f performs relaxed concept erasure for $\epsilon(u) > 0$, if the utility is $I(Z; X) \geq u$ and $I(Z; A) = \epsilon(u)$, where $\epsilon(u) = \inf_f \{I(Z; A) | I(Z; X) \geq u, A \rightarrow X \xrightarrow{f} Z\}$.*

In Definition 3.5, $\epsilon(u)$ is the minimum achievable privacy leakage, $I(Z; A)$, such that the utility is at least

¹This is often termed as privacy funnel, but we will refer to it as erasure funnel in the context of concept erasure.

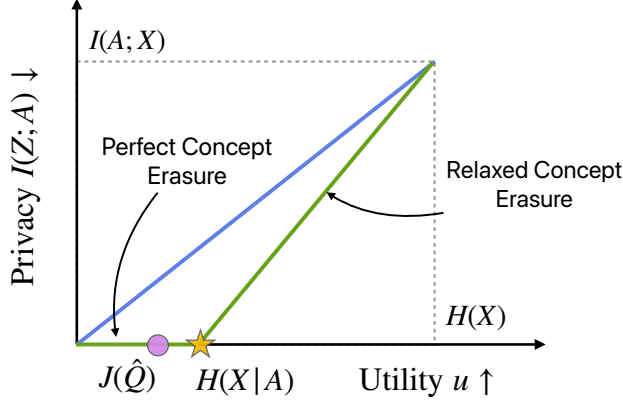


Figure 1: Schematic diagram of the privacy-utility tradeoff and illustration of the bounds of erasure funnel function, $\epsilon(u)$ (Calmon et al., 2017). The region shown by the green line is desirable for concept erasure as it is the minimum $I(Z; A)$ for a given utility $I(Z; X) \geq u$.

$I(Z; X) \geq u$. This definition considers the utility, $I(Z; X)$, and focuses on finding an erasure function f that does not erase all information. This is useful in applications such as fairness, where it is not necessary to completely erase concepts to achieve fair outcomes.

3.3 Outer Bounds For Concept Erasure

We build on the definitions introduced in the previous section and present the outer bounds for privacy. Ideally, we want to *minimize* privacy leakage, $I(Z; A)$ while *maximizing* utility, $I(Z; X)$. Here, we utilize the erasure funnel (Def. 3.4) to answer these questions. The erasure funnel extends beyond relaxed erasure and can offer insights into perfect erasure ($\epsilon(u) = 0$). We will use $\epsilon(u)$ to answer the following research questions:

- (RQ1) What is the maximum utility u during perfect concept erasure (when $\epsilon = 0$)?
- (RQ2) What is the outer bound of privacy $\epsilon(u)$ (minimum $\epsilon(u)$) during relaxed concept erasure?

As obtaining the precise analytic expression for $\epsilon(u)$ is difficult, we present its bounds (Calmon et al., 2017) to answer the above research questions.

Lemma 3.6 (Outer Bounds for $\epsilon(u)$ (Calmon et al., 2017)). *For $0 \leq u \leq H(X)$, erasure funnel function, $\epsilon(u)$, is bounded as:*

$$\max\{0, u - H(X|A)\} \leq \epsilon(u) \leq \frac{uI(A; X)}{H(X)}.$$

In Figure 1, we plot the result of Lemma 3.6, where the y -axis indicates the privacy $I(Z; A)$, and the x -axis indicates the utility, u . We observe that $\epsilon(u)$ lies within a funnel-like structure with upper bound shown in blue and lower bound in green. In this

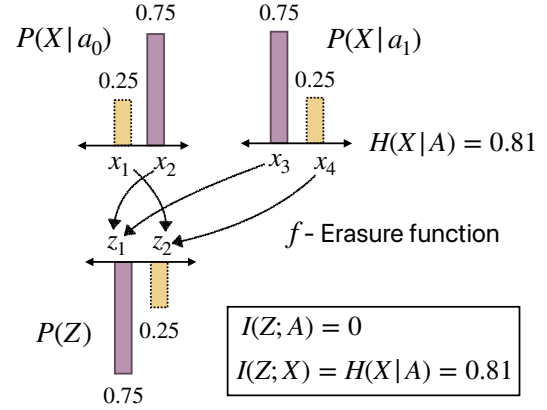


Figure 2: An illustration of a perfect erasure function. The input representations of each group are sampled from 2D distributions. The erasure function f in the figure achieves the outer bounds for concept erasure.

plot, perfect concept erasure is the flat portion of the green line where $\epsilon(u) = 0$. During perfect erasure, we observe that the maximum possible utility is $H(X|A)$ shown by $\star$. This is the same result we derived in Lemma 3.3 and provides the solution for (RQ1). During relaxed concept erasure $\epsilon(u) > 0$, the minimum mutual information is $\epsilon(u) = \frac{uI(A; X)}{H(X)}$, which is a linear function of the utility, u . This provides the solution for (RQ2). We will discuss the importance of $\bullet$ in Section 4.

We would like to emphasize that these outer bounds might not be achievable for any random variables X and A . Therefore, our next focus will be on understanding the data constraints under which the outer bounds can be attained and identifying the erasure functions that achieve them. In this work, we only consider perfect concept erasure and its corresponding erasure functions. We defer the derivation of erasure functions that achieve relaxed concept erasure to future works.

4 PERFECT ERASURE FUNCTIONS (PEF)

In this section, we present an algorithm for concept erasure, which is inspired by achieving perfect concept erasure, $I(Z; A) = 0$, along with the utility outer bound, $I(Z; X) = H(X|A)$ ($\star$ in Fig. 1). We will refer to these methods as *perfect erasure functions*.

To motivate our method, we consider a simple example in Fig. 2. We consider a scenario with two concept groups, $\mathcal{A} = \{a_0, a_1\}$ with equal prior $p(a_0) = p(a_1) = 0.5$. We consider the data representation space $\mathcal{X}$ to be $\{x_1, x_2, x_3, x_4\}$. Each group has a two dimensional representation – $P(X|A = a_0)$ has support of $\{x_1, x_2\}$ and $P(X|A = a_1)$ has a support of $\{x_3, x_4\}$.

In this example, there is an erasure function, f , which